

Linear Algebra

Solutions to Practice Problems

Chapter 3 — Finite Dimensional Vector Spaces

How This Pairs With Your Notes

This is the companion answer key to *LinearAlgebra_Ch3_Notes.pdf*. Each problem is restated in an orange box, followed by a full worked solution in teal. Try each problem yourself first.

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1 Solutions – §3.1 (Vector Spaces & Subspaces)

Problem 3.1.1

Verify that $P_2(\mathbb{R})$ is closed under addition and scalar multiplication, using two polynomials of your choice.

Solution

Take $p(x) = 1 + 2x - x^2$ and $q(x) = 3 - x + 4x^2$, both degree ≤ 2 . Their sum $p(x) + q(x) = 4 + x + 3x^2$ is still degree ≤ 2 , so it's still in $P_2(\mathbb{R})$. For a scalar, say 5: $5p(x) = 5 + 10x - 5x^2$, again degree ≤ 2 . In general, adding two polynomials of degree ≤ 2 or scaling one can never *raise* the degree above 2 (it can only lower it, if leading terms cancel) – so $P_2(\mathbb{R})$ is closed under both operations.

Problem 3.1.2

Is $S = \{(x, y, z) \in \mathbb{R}^3 : x + 2y - z = 0\}$ a subspace of $\mathbb{R}^3$?

Solution

Yes. Take $u = (x_1, y_1, z_1), v = (x_2, y_2, z_2) \in S$ (so $x_1 + 2y_1 - z_1 = 0$ and $x_2 + 2y_2 - z_2 = 0$) and scalars α, β . Then $\alpha u + \beta v = (\alpha x_1 + \beta x_2, \alpha y_1 + \beta y_2, \alpha z_1 + \beta z_2)$, and checking the defining condition:

$$(\alpha x_1 + \beta x_2) + 2(\alpha y_1 + \beta y_2) - (\alpha z_1 + \beta z_2) = \alpha(x_1 + 2y_1 - z_1) + \beta(x_2 + 2y_2 - z_2) = \alpha(0) + \beta(0) = 0.$$

So $\alpha u + \beta v \in S$: closed under linear combinations, hence a subspace (Definition 3.1.6).

Problem 3.1.3

Is $S = \{(x, y, z) \in \mathbb{R}^3 : x + 2y - z = 1\}$ a subspace? Explain the difference from Problem 3.1.2.

Solution

No. The origin $(0, 0, 0)$ must belong to every subspace (since $0 \cdot u = 0$ for any u already in it), but $0 + 2(0) - 0 = 0 \neq 1$, so $(0, 0, 0) \notin S$. The difference from Problem 3.1.2 is the right-hand side: $= 0$ describes a plane *through* the origin (a subspace), while $= 1$ describes a plane that *misses* the origin (not a subspace) – exactly the geometric picture from the notes.

Problem 3.1.4

Find $L(S)$ for $S = \{(1, 0, 1), (1, 1, 0)\} \subset \mathbb{R}^3$.

Solution

A general element is $a(1, 0, 1) + b(1, 1, 0) = (a + b, b, a)$. Writing $x = a + b$, $y = b$, $z = a$: then $b = y$ and $a = z$, so $x = a + b = z + y$, i.e. $x - y - z = 0$. So

$$L(S) = \{(x, y, z) \in \mathbb{R}^3 : x - y - z = 0\}.$$

(Quick check: $(1, 0, 1)$ gives $1 - 0 - 1 = 0$ ✓; $(1, 1, 0)$ gives $1 - 1 - 0 = 0$ ✓.)

Problem 3.1.5

Find a basis for the null space of $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{pmatrix}$.

Solution

Row-reduce: $R_{21}(-1)$ gives $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}$, then $R_{12}(-1)$ gives $\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \end{pmatrix}$ (RREF). Reading off $Ax = 0$: $x_1 - x_3 = 0$ and $x_2 + 2x_3 = 0$, with x_3 free. Setting $x_3 = t$: $x_1 = t$, $x_2 = -2t$, $x_3 = t$, i.e.

$$N(A) = L(\{(1, -2, 1)\}).$$

Check: $A(1, -2, 1)^t = (1 - 2 + 1, 1 - 4 + 3)^t = (0, 0)^t$. ✓

Problem 3.1.6 (Challenge)

Show that the set of all 2×2 symmetric matrices is a subspace of $M_{2 \times 2}(\mathbb{R})$.

Solution

Let $S = \{A \in M_{2 \times 2}(\mathbb{R}) : A^t = A\}$. Take $A, B \in S$ (so $A^t = A$, $B^t = B$) and scalars α, β . Using the transpose properties from Chapter 1, $(\alpha A + \beta B)^t = \alpha A^t + \beta B^t = \alpha A + \beta B$. So $\alpha A + \beta B$ is itself symmetric, i.e. $\alpha A + \beta B \in S$. Closed under linear combinations $\Rightarrow S$ is a subspace.

2 Solutions – §3.2 (Linear Independence)**Problem 3.2.1**

Are $(1, 2), (2, 4)$ linearly independent in $\mathbb{R}^2$?

Solution

No. $(2, 4) = 2 \times (1, 2)$, so $2(1, 2) - 1(2, 4) = (0, 0)$ is a *nontrivial* combination (coefficients 2, -1 , not both zero) equal to the zero vector. By Definition 3.2.1, this makes the set **dependent**.

Problem 3.2.2

Are $(1, 0, 0)$, $(1, 1, 0)$, $(1, 1, 1)$ linearly independent in $\mathbb{R}^3$?

Solution

Stack as rows and compute the determinant:

$$\det \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} = 1(1 - 0) - 0 + 0 = 1 \neq 0.$$

A nonzero determinant means the only solution to $\alpha_1(1, 0, 0) + \alpha_2(1, 1, 0) + \alpha_3(1, 1, 1) = 0$ is $\alpha_1 = \alpha_2 = \alpha_3 = 0$. **The set is independent.**

Problem 3.2.3

Can a set containing the zero vector ever be linearly independent?

Solution

No, by Proposition 3.2.3(1). Directly: if $0 \in S$, pick any nonzero scalar γ for the coefficient of 0 and 0 for every other vector's coefficient. This gives $\gamma \cdot 0 + 0 \cdot (\text{everything else}) = 0$ – a *nontrivial* combination (since $\gamma \neq 0$) equal to zero. So any set containing the zero vector is automatically dependent.

Problem 3.2.4

If $\{u, v, w\}$ is independent, is $\{u, v\}$ independent? What about $\{u, v, w, u + v\}$?

Solution

$\{u, v\}$: **yes**, independent – any subset of an independent set is independent (Proposition 3.2.3(2)).
 $\{u, v, w, u + v\}$: **no**, dependent – the combination $1 \cdot (u + v) - 1 \cdot u - 1 \cdot v + 0 \cdot w = 0$ is nontrivial (the coefficients 1, -1 , -1 , 0 are not all zero) and equals the zero vector, so the enlarged set is dependent. (This matches Theorem 3.2.4: $u + v$ is already in the span of u, v , so adding it creates redundancy.)

Problem 3.2.5 (Challenge)

Suppose $\{v_1, v_2\}$ is independent and $v_3 = 2v_1 - v_2$. Explain why $\{v_1, v_2, v_3\}$ must be dependent.

Solution

By definition, $v_3 \in L(v_1, v_2)$ (it's built as a combination of v_1 and v_2). Theorem 3.2.4 says: if $\{v_1, v_2\}$ is independent and adding v_3 keeps it independent, then v_3 would have to be *outside* $L(v_1, v_2)$ – but $v_3 = 2v_1 - v_2$ clearly lies inside it. So the “stays independent” option is impossible, and $\{v_1, v_2, v_3\}$ **must be dependent** – with no further computation needed, exactly as the problem asks. (Explicitly: $2v_1 - v_2 - v_3 = 0$ is a nontrivial relation.)

3 Solutions – §3.3 (Bases & Dimension)

Problem 3.3.1

State the standard basis of $\mathbb{R}^4$ and give $\dim(\mathbb{R}^4)$.

Solution

Standard basis: $e_1 = (1, 0, 0, 0)$, $e_2 = (0, 1, 0, 0)$, $e_3 = (0, 0, 1, 0)$, $e_4 = (0, 0, 0, 1)$. Since this basis has 4 vectors, $\dim(\mathbb{R}^4) = 4$.

Problem 3.3.2

What is $\dim(P_3(\mathbb{R}))$? Name a basis.

Solution

Basis: $\{1, x, x^2, x^3\}$ (every polynomial of degree ≤ 3 is a unique combination of these four). So $\dim(P_3(\mathbb{R})) = 4$ (in general, $\dim(P_n(\mathbb{R})) = n + 1$ – easy to forget the “+1” from the constant term).

Problem 3.3.3

Is $\{(1, 1), (2, 2)\}$ a basis of $\mathbb{R}^2$?

Solution

No. $(2, 2) = 2 \times (1, 1)$, so the set is linearly *dependent* (same reasoning as Problem 3.2.1) – and a basis must be independent. (Also, both vectors point along the same line, so their span is only a 1-dimensional line, not all of $\mathbb{R}^2$ – failing the spanning condition too.)

Problem 3.3.4

Extend $\{(1, 1, 0)\}$ to a basis of $\mathbb{R}^3$.

Solution

Add, e.g., $(0, 1, 0)$ and $(0, 0, 1)$: check independence of $\{(1, 1, 0), (0, 1, 0), (0, 0, 1)\}$ via the determinant

$$\det \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = 1(1 - 0) - 1(0 - 0) + 0 = 1 \neq 0,$$

confirming independence. Three independent vectors in a 3-dimensional space automatically span it too (Corollary 3.3.16), so $\{(1, 1, 0), (0, 1, 0), (0, 0, 1)\}$ is a valid basis of $\mathbb{R}^3$.

Problem 3.3.5 (Challenge)

For $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}$, find $\text{rank}(A)$ and verify Proposition 3.3.20 by finding the dimensions of the row space and column space separately.

Solution

Row 2 = 2 × row 1, so row-reducing leaves one nonzero row: $\text{rank}(A) = 1$. **Row space:** spanned by $(1, 2, 3)$ alone (row 2 is redundant), so $\dim(\text{RowSpace}) = 1$. **Column space:** the three columns are $(1, 2)^t, (2, 4)^t, (3, 6)^t$ – every one of them is a multiple of $(1, 2)^t$, so they span only a 1-dimensional space: $\dim(\text{ColumnSpace}) = 1$. Both dimensions equal 1, matching $\text{rank}(A) = 1$.

and confirming Proposition 3.3.20.

4 Solutions – §3.4 (Ordered Bases & Coordinates)

Problem 3.4.1

Find the coordinates of $(3, 5)$ with respect to $B = ((1, 1), (1, -1))$.

Solution

Solve $a(1, 1) + b(1, -1) = (3, 5)$: $a + b = 3$ and $a - b = 5$. Adding: $2a = 8 \Rightarrow a = 4$; then $b = 3 - 4 = -1$. So

$$[(3, 5)]_B = (4, -1)^t.$$

Check: $4(1, 1) + (-1)(1, -1) = (4 - 1, 4 + 1) = (3, 5)$. ✓

Problem 3.4.2

Find the coordinates of $(1, 2, 3)$ with respect to the standard ordered basis of $\mathbb{R}^3$.

Solution

$[(1, 2, 3)]_{\text{standard}} = (1, 2, 3)^t$ – **immediate**, because $(1, 2, 3) = 1 \cdot e_1 + 2 \cdot e_2 + 3 \cdot e_3$ by definition of the standard basis: a vector’s components *are* its standard-basis coordinates. This is exactly why the standard basis feels “invisible” in everyday use – coordinates and components coincide only for this one special basis.

Problem 3.4.3

If $[v]_B = (2, -1, 3)^t$ for ordered basis $B = (u_1, u_2, u_3)$, write v explicitly.

Solution

By Definition 3.4.3, the coordinate vector lists the coefficients in order:

$$v = 2u_1 - u_2 + 3u_3.$$

Problem 3.4.4

Explain why the same vector can have different-looking coordinates under different ordered bases.

Solution

Coordinates are not a property of the vector alone – they describe *how that vector is built* out of a specific set of building blocks (the chosen basis). Change the building blocks, and the “recipe” for the same finished vector changes too, even though the vector itself never moves. It’s exactly like giving directions to the same building using two different street grids: the address numbers differ completely, but the building is the same physical place. Problem 3.4.1 illustrates this directly: $(3, 5)$ is $(3, 5)^t$ in the standard basis but $(4, -1)^t$ in B .

Problem 3.4.5 (Challenge)

Find the change-of-basis matrix A with $[x]_{B_1} = A[x]_{B_2}$ for $B_1 = ((1, 0), (0, 1))$ and $B_2 = ((1, 1), (1, -1))$, and use it to recompute Problem 3.4.1.

Solution

By Theorem 3.4.5, the columns of A are the B_2 basis vectors written in B_1 (standard) coordinates – but since B_1 is standard, that’s just the vectors themselves:

$$A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

To recompute Problem 3.4.1 ($[v]_{B_1} = (3, 5)^t$ known, want $[v]_{B_2}$): rearrange $[x]_{B_1} = A[x]_{B_2}$ to $[x]_{B_2} = A^{-1}[x]_{B_1}$. Using the 2×2 inverse shortcut, $\det(A) = -2$, so $A^{-1} = \frac{1}{-2} \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix}$. Then

$$[v]_{B_2} = A^{-1} \begin{pmatrix} 3 \\ 5 \end{pmatrix} = \begin{pmatrix} 1/2(3) + 1/2(5) \\ 1/2(3) - 1/2(5) \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \end{pmatrix},$$

exactly matching $(4, -1)^t$ from Problem 3.4.1 – the change-of-basis matrix reproduces the direct computation perfectly.

That covers every practice problem for Chapter 3. Ask for Chapter 4’s notes and solutions whenever you’re ready to continue.